

Physics-aware layout representations for transferable models of planar superconducting devices

Author list to be finalized

Abstract

Electromagnetic simulations for quantum-device design are costly, while the resulting datasets remain fragmented across component generators, parameter dictionaries, and design tools. We develop a deterministic representation that maps planar GDS geometry, ground, conductor connectivity, and arbitrary terminal sets into a common physics-aware space. We first evaluate the representation itself: nuisance invariance, sensitivity and smoothness under physical changes, feature-block contributions, component-family structure, and preservation of galvanic topology. We then test whether representation distance carries capacitance information, whether a shared model improves on native tabular baselines, and whether source-family training reduces the target simulations needed for unseen families and composite multi-terminal systems. The arbitrary-terminal v4 views passed all nine frozen planar numerical gates. Across three families, their mean within-family R^2 values in $\log(1 + C)$ were 0.987, 0.986, and 0.978. Representation distance was only modestly associated with capacitance difference ($\rho = 0.251$), and historical cross-family transfer did not consistently outperform target-only fitting. Training-family diversity nevertheless improved a matched-budget generalist, while a three-terminal composite established structural—but not yet high-fidelity predictive—zero-shot generalization. The planar interface is designed to accept a future material-stack sidecar, but multi-plane and flip-chip generalization are outside the present validation scope.

Introduction

High-fidelity electromagnetic simulation is a bottleneck in superconducting quantum-hardware design. Individual groups amortize this cost through parameter sweeps, interpolation, and surrogate models, but each table is normally tied to one component generator. A parameter such as `finger_length` in an interdigital capacitor has no literal counterpart in a transverse-cross device, even when both layouts pose related electrostatic questions. The incompatibility becomes most costly when a new topology requires another large simulation campaign.

GDS is a more portable interface than the code that generated a component, but geometry alone is not yet a scientific representation. A useful encoder must distinguish physical scale from drawing pose, identify ground and galvanically connected conductors, preserve terminal identity without depending on arbitrary terminal ordering, and expose the correct structure of a variable-size capacitance matrix. It should detect physically important changes at both small and large scales without responding strongly to irrelevant translations or rotations.

Here we ask three questions. First, can a static planar GDS embedding satisfy a predeclared physical and topological contract across heterogeneous Qiskit Metal and SQuADS components? Second, does the resulting geometry carry enough capacitance information to outperform family-specific tabular representations and predict which families can transfer useful knowledge? Third, can a generalist representation recognize a newly composed multi-terminal system and adapt with a small number of new simulations? We treat all three as empirical questions and separate representation acceptance from downstream accuracy.

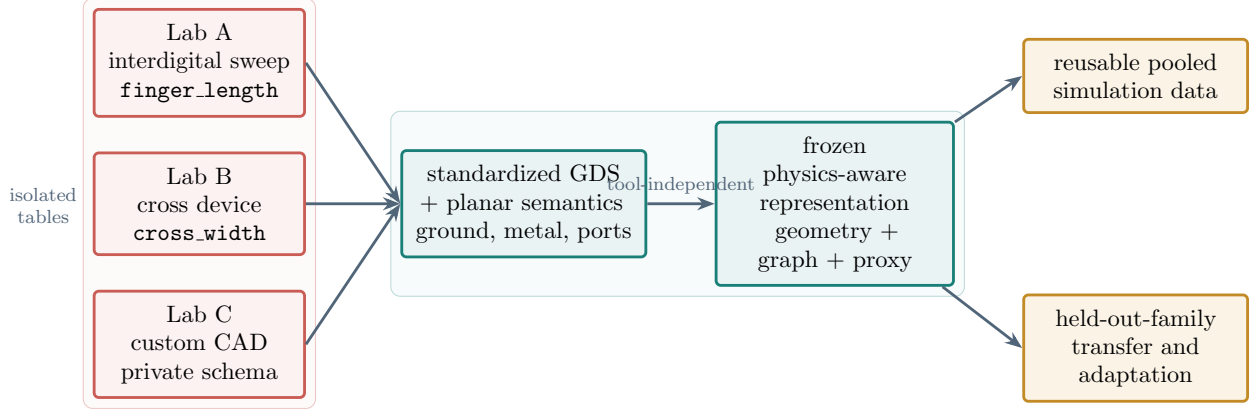


Figure 1: From fragmented tables to a shared layout interface. Native parameter dictionaries remain useful within a generator but do not define common features across families. Standardized planar GDS and explicit semantics provide the input to one frozen representation. The downstream reuse and transfer arrows state the hypotheses tested in this paper; they are not assumed consequences of the file format.

Results

A static planar representation with an explicit physical contract

The encoder begins with a planar GDS semantics contract for signal metal, ground, terminals, and connectivity. Polygon processing produces connected conductor bodies and their relationships to local ground. Geometry, physical scale, proximity, topology, and a deterministic low-fidelity capacitance proxy are recorded as identifiable feature blocks. The multi-terminal interface represents an arbitrary conductor/terminal set and supports a matrix-valued capacitance target rather than reducing every device to one ordered terminal pair.

Translation, in-plane rotation, reflection, and irrelevant outer-domain crop are treated as nuisance transformations. Uniform scale, gap, overlap, width, facing length, ground clearance, body count, terminal configuration, and galvanic joins are physical transformations. These assignments are frozen before aggregate scores are calculated.

In the first frozen planar acceptance run, the v4 layout, graph-pair, and hybrid-pair views each passed all nine numerical gates; the v2 and conditioned v2 controls passed five of nine. At an arbitrary 37° rotation, relative drift was 9×10^{-6} for the v4 layout view, 1×10^{-5} for the graph-pair view, and 8.06×10^{-4} for the hybrid view, all below the predeclared 3×10^{-3} threshold. These are numerical tests on controlled GDS counterfactuals, not evidence for multi-plane invariance.

The v4 layout view pools a 64-coordinate global block, three 64-coordinate statistics of conductor-node features, and three 96-coordinate statistics of pair-edge features. The reciprocal graph-pair view concatenates global, endpoint-sum, endpoint-difference, selected-edge, and all-other-conductor context blocks. The production hybrid pair view combines the 240-coordinate v3 fixed-stack pair core with 192 coordinates of all-other-conductor context.

Small and large physical changes produce detectable, smooth signals

We generate paired narrow- and broad-range sweeps over representative physical parameters. For each path we measure full-vector displacement, block-level displacement, local finite differences, and discontinuities near any histogram or basis boundaries. Pose controls are run on the same layouts.

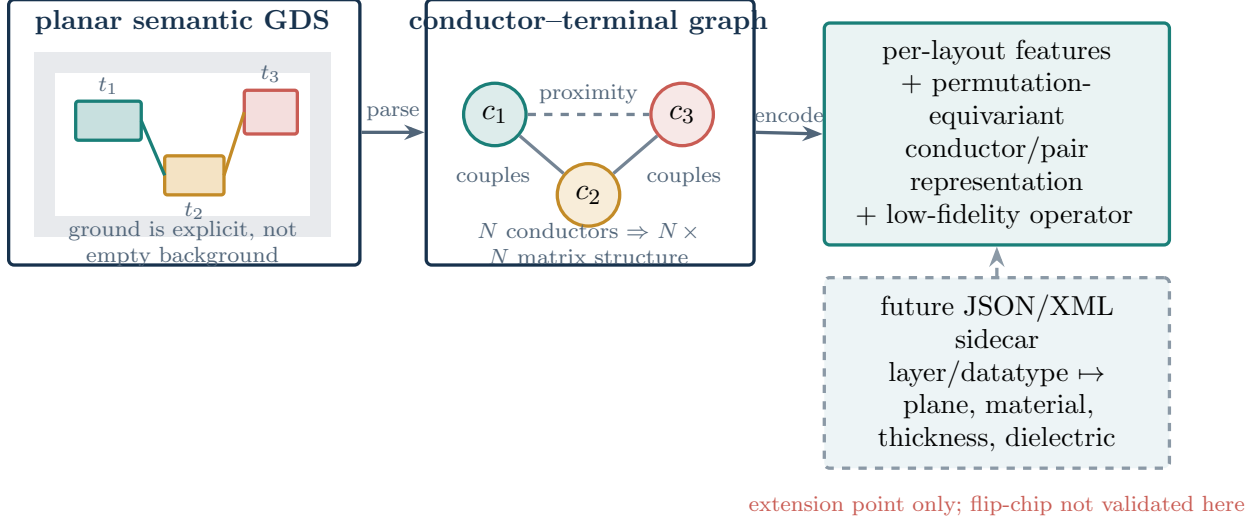


Figure 2: Planar arbitrary-terminal and future stack contract. The planar GDS parser identifies ground, connected conductor bodies, and attached terminals; the graph determines the matrix structure and must transform equivariantly when terminal labels are permuted. A versioned material-stack sidecar is reserved as an additive input for later multi-plane work. No stack-awareness result is claimed in the present planar study.

Cumulative and drop-one-block ablations compare v2 and later candidates and identify whether each block contributes desired sensitivity rather than nuisance leakage.

All v4 views responded smoothly to the four controlled paths and preserved perfect rank ordering along the terminal-gap path. In the layout-view cumulative and drop-one ablations, the physical-to-noise response ratio remained between 9.4×10^3 and 1.13×10^4 , so no single retained pooled block was solely responsible for the invariance result.

A component atlas tests breadth and structured sampling

We instantiate and export one valid planar GDS for every feasible Qiskit Metal component. Every attempted component remains in the audit table: unsupported, invalid, or semantically ambiguous exports are exclusions to explain rather than points to hide. We test uniqueness in the full representation and use a two-dimensional projection only for visualization; separation in a projection is not evidence of vector uniqueness.

Dense **GeneralizedCapNInterdigital** sweeps provide the principal within-family example. The newly supplied sweep families and relevant **SQuADDS** components test whether families occupy organized neighborhoods and whether parameter sweeps trace reproducible trajectories through those neighborhoods.

The frozen Qiskit Metal inventory contained 38 production classes. Thirty-six could be exported as valid standardized planar GDS, while **OpenToGround** and **ShortToGround** were retained in the manifest as semantic/context-only entries rather than manufactured geometries. The 36 valid exports formed 31 independently measured physical-geometry equivalence classes, and the v4 layout view produced exactly 31 vectors. Thus every observed vector equality was explained by identical physical geometry. In a separate deterministic sample of 144 supplied layouts (24 from each of six campaigns), all 144 v4 layout vectors were distinct.

The exact-vector audit therefore found no unexplained collision in either the Qiskit Metal

Planar embedding acceptance scorecard

Representation	v2	PASS	PASS	FAIL	FAIL	PASS	FAIL	FAIL	PASS	PASS	FAIL
	v2 conditioned	PASS	PASS	FAIL	FAIL	PASS	FAIL	FAIL	PASS	PASS	FAIL
	v3 core	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	FAIL
	v3 complete	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	FAIL
	v4 layout	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
	v4 graph pair	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
	v4 hybrid pair	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
		determinism	translation	rotation 90	rotation 37	reflection	outer crop	sensitivity ratio	continuity	gap ordering	arbitrary terminals
Predeclared acceptance requirement											

Figure 3: Frozen planar representation acceptance scorecard. The text reports the established nine-gate totals and arbitrary-rotation drift values. This panel exposes each gate and representation view separately so an aggregate pass count cannot hide a failed physical contract.

singleton inventory or the sampled supplied campaigns. The two-dimensional atlas is retained only as an interpretable visualization of family structure; it is not used to certify uniqueness.

Planar composition preserves ports and galvanic topology

The recordings motivate both representation differences and modular composition. We therefore test differences along controlled parameter paths and compare a port-aware composition of module representations with the embedding computed directly from the joined GDS. Composition is not assumed to be raw vector addition: relative position, scale, ports, shared ground, and conductor merges must be supplied to the operation. A decisive example joins two two-conductor modules so that one galvanic merge produces a three-conductor system and a correspondingly different capacitance-matrix shape. Higher terminal counts test that the interface is not specialized to this example.

Naive addition of the two 544-coordinate layout views had high cosine similarity (0.991) but a relative error of 0.869 against the independently encoded joined GDS, decisively rejecting raw addition as a composition law. Merging the shared port/net in geometry space produced the correct three-conductor graph and a 4×4 conductor-plus-ground proxy. Pose-matched physical difference vectors agreed with cosine 0.99997 and relative error 0.0091; the wider-gap control’s 37° rotation drift was 9.87×10^{-4} .

A common representation is compared with native parameter tables

Embedding and native CSV/parameter-dictionary predictors receive identical simulation labels, train/test identities, preprocessing budgets, and comparable model capacity. We report in-family performance and then hold out entire families. Because native schemas differ, their inability to form a common input is documented explicitly rather than concealed by an arbitrary hand-aligned table.

On repeated design-disjoint within-family splits, v4 reached mean R^2 values of 0.987, 0.986, and 0.978 for generalized IDC, Tee, and transverse-cross families. Native options reached 0.969, 0.940, and 0.981, respectively. Thus the universal geometry view retained or improved within-family

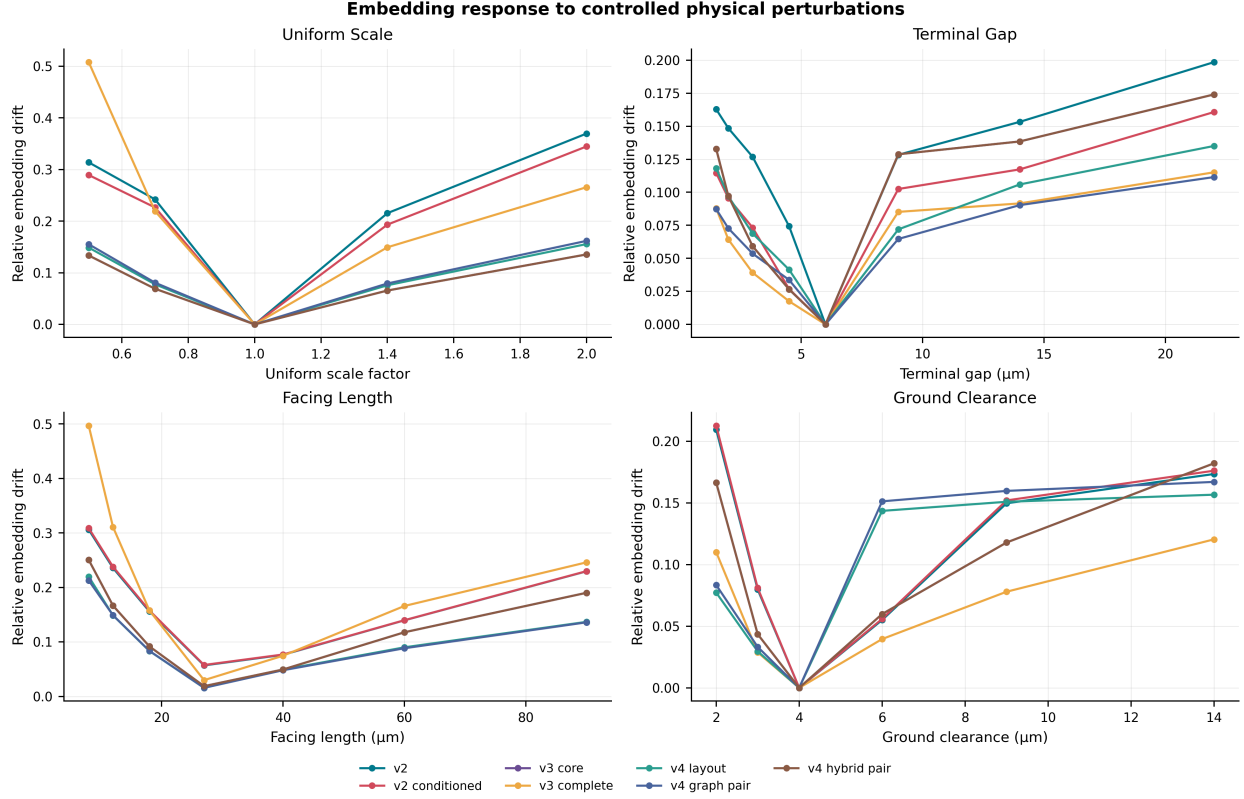


Figure 4: Controlled physical trajectories at narrow and broad scales. The panels compare parameter change with full-vector displacement. The v4 views leave the baseline smoothly for uniform scale, terminal gap, facing length, and local ground clearance while maintaining the frozen continuity and gap-ordering gates. Monotonicity is interpreted only where a physical direction was declared in advance.

performance for the two capacitors, whereas the compact native Transmon coordinates remained slightly stronger for the cross and had lower absolute error.

As a provisional downstream audit using one frozen random-feature ridge head and strict leave-one-family-out splits (894 designs per family), v4 obtained R^2 values of 0.970, 0.743, and 0.727 for held-out `CapNInterdigitalTee`, `GeneralizedCapNInterdigital`, and `TransmonCross`, respectively. The corresponding RMSE values were 3.107, 1.002, and 1.331 fF, and median absolute errors were 1.060, 0.635, and 0.787 fF. v2 collapsed on the held-out transverse cross ($R^2 = -2.494$; RMSE 6.739 fF; median absolute error 5.119 fF). The v3 core remained slightly better than v4 on that target ($R^2 = 0.792$), so repeated-seed and model-class controls are still required before claiming v4 dominance.

Representation proximity is calibrated against capacitance

We first compute a family-balanced similarity matrix and predeclare how unequal family sizes are weighted. The nearest and most distant family pairs define two transfer case studies. Across the full evaluation set, representation distance is compared with normalized capacitance difference using design-cluster uncertainty intervals. Similarity is treated as a hypothesis about transfer, not a physical result by itself.

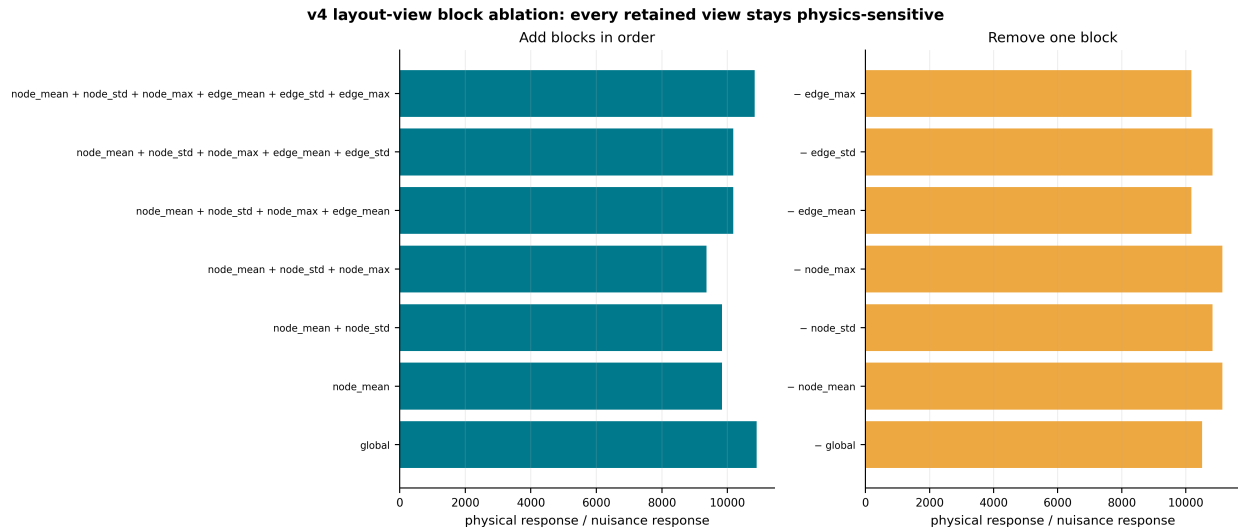


Figure 5: Cumulative and drop-one-block ablation of v2 and later planar representations. Each block is evaluated against both nuisance controls and physical perturbations; downstream accuracy alone is not an acceptance test. Every cumulative and drop-one v4 layout view retained a physical-to-nuisance response ratio above 9.4×10^3 .

Across 12,000 family-balanced design pairs, Spearman association between embedding distance and normalized capacitance difference was 0.082 for v2, 0.255 for the v3 fixed-stack core, and 0.251 for the v4 pair view. Design-cluster bootstrap 95% intervals were [0.073, 0.123], [0.237, 0.296], and [0.235, 0.294], respectively. The positive but modest values support geometry distance as a retrieval prior, not as a calibrated capacitance metric.

Near- and far-family transfer is measured by target-label efficiency

For both the high-similarity and low-similarity source–target family pairs, a source specialist is adapted using nested target subsets of 1, 5, 10, 50, and 100 simulations. The target test set remains fixed. Source-initialized models are compared with target-only and no-transfer baselines over repeated seeds. The primary endpoint is the number of target simulations needed to reach a predeclared matrix-error tolerance; conventional R^2 , RMSE, and MAE are secondary summaries.

The repeated learning curves did not show a universal source-transfer advantage. At 100 target labels, target-only v4 achieved mean $R^2 = 0.943$, RMSE 1.936 fF, and median absolute error 0.584 fF, compared with $R^2 = 0.911$, RMSE 2.408 fF, and median absolute error 0.765 fF for pooled source-plus-target training. One-label estimates were unstable and negative in mean R^2 for all methods. Near/far geometry-centroid selection therefore did not overcome the solver-campaign and topology confounding in these historical tables.

Controlled diversity tests a generalist model

Generalist training sets grow in a controlled sequence: more samples within a family, nearby families, distant families, and mixtures that expand coverage of representation space. Sample-count-matched controls separate diversity from dataset size. Every model is evaluated on the same validation sets, including a family absent from training. Leave-one-cluster-out, within-cluster interpolation, nearby extrapolation, cluster-edge, and heavy-tail analyses test whether the model learned local structure

or collapsed toward a global mean.

Matched-budget family diversity was consistently more useful than adding rows from only one topology. With 600 training designs drawn from all three families, mean cross-task R^2 was 0.881 for v2, 0.942 for v3, and 0.941 for v4. In contrast, 600 designs from one family gave mean R^2 values of -1.091 , -0.601 , and -0.356 , respectively, because unseen-family predictions remained poor.

An unseen composite system closes the workflow

The final structural experiment connects previously modeled planar modules into a multi-terminal component absent from training. Before any predictive claim, the pipeline must identify the new conductor graph and requested capacitance-matrix shape. A subsequent high-fidelity composite sweep and target-only adaptation control are intentionally deferred until converged solver labels exist.

The unseen joined GDS was parsed as three signal conductors with no unmarked island and produced the expected 4×4 symmetric conductor-plus-ground low-fidelity operator. This is a structural zero-shot pass only. Because no converged three-dimensional capacitance matrix or composite sweep is currently available, zero-shot accuracy and adaptation curves are explicitly untested.

Discussion

The intended contribution is an auditable bridge between otherwise incompatible simulation datasets, not a claim that a learned surrogate replaces electromagnetic simulation. The representation tests establish whether its coordinates behave physically; held-out-family and composite experiments then measure whether that prior reduces the need for new high-fidelity labels.

The current validation scope is planar. The software interface should reserve a versioned JSON or XML mapping from GDS layer/datatype to plane, material, thickness, and dielectric environment. Flip-chip claims require crossed geometry-stack sweeps, same-projection/different-plane controls, and held-out stack tests, and are deferred until those data exist. Likewise, “foundation model,” universal composition, and transfer to semiconductor or photonic devices remain future claims unless directly demonstrated.

Methods

Planar GDS semantics and validation

The exact layer/datatype assignments are versioned with the dataset rather than hard-coded in the scientific claim. The schema identifies signal and ground regions, named or ordered terminals, exclusions/etch, and connectivity rules. Before embedding, every file is checked for valid geometry, connected-component structure, terminal attachment, and consistency between terminal/conductor count and the requested capacitance matrix.

Representation acceptance metrics

For vectors $e(x)$ and paired transformations T , relative drift is

$$d_T = \frac{\|e(Tx) - e(x)\|_2}{\max(\|e(x)\|_2, \epsilon)}. \tag{1}$$

Nuisance transformations are required to remain below frozen numerical tolerances. Physical paths must be non-degenerate and locally smooth; monotonic ordering is required only for features and controls for which the physical direction is justified. We additionally report blockwise effects so aggregate normalization cannot mask a failed block.

For arbitrary terminal sets, relabeling tests require the conductor/terminal representation and predicted matrix to transform equivariantly under the same permutation. Matrix errors are reported per independent entry and with a predeclared matrix-level aggregation.

Predictive and transfer evaluation

All preprocessing is fitted on training data only. Grouped splits use stable design identities so duplicate geometry cannot cross train and test sets. Held-out-family tests, nested target-data learning curves, and diversity ablations use fixed test sets, repeated seeds, and uncertainty intervals. Family-pair selection is based on training-side representation statistics, without consulting held-out target labels.

Data and code availability

Code, GDS generation recipes, semantic schemas, acceptance thresholds, executed notebooks, and exact exclusion logs will be released with immutable versions after review. No external repository or dataset hub is modified by this working draft.

References

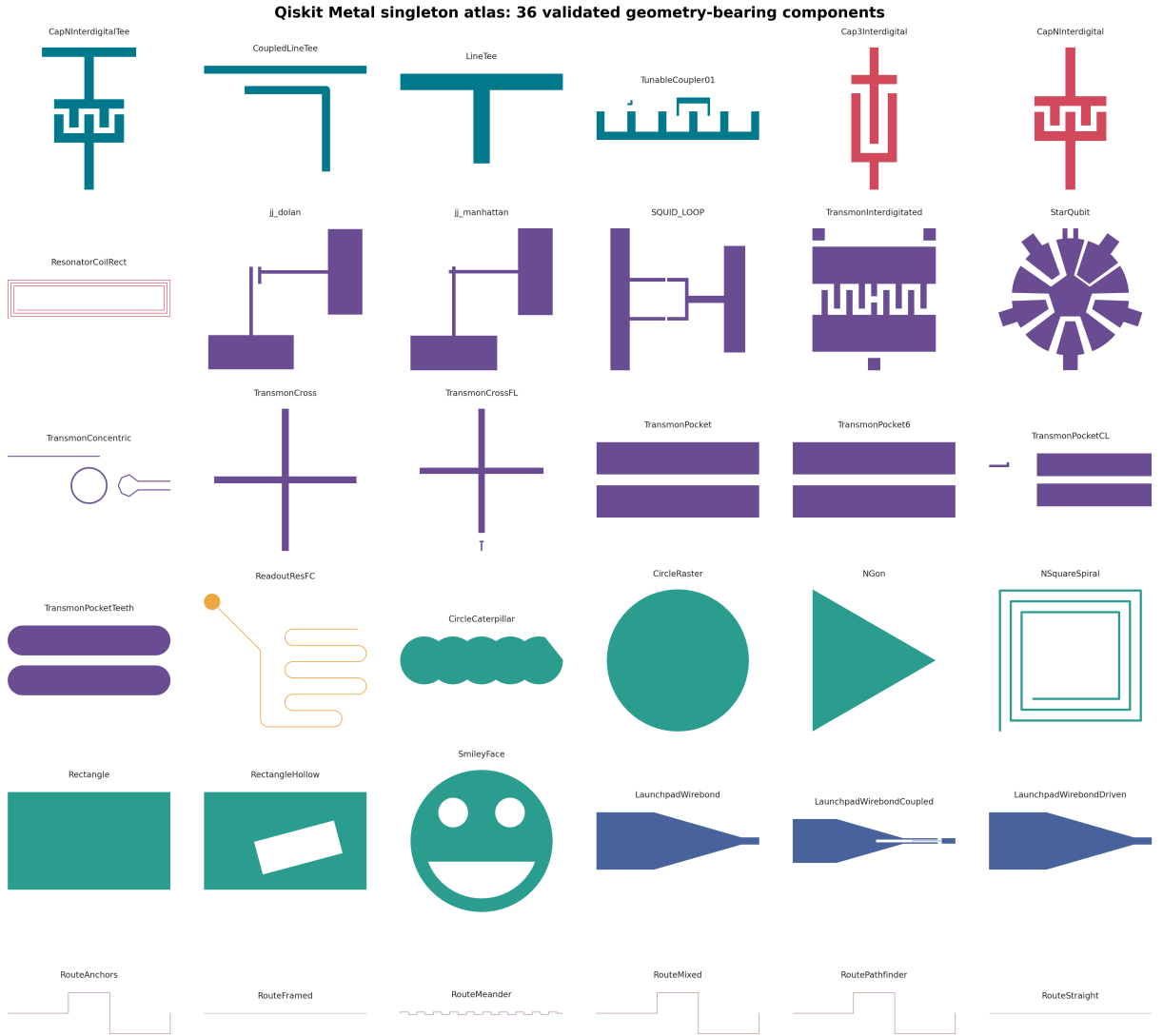


Figure 6: Qiskit Metal singleton atlas and structured sweep coverage. Of 38 production classes, 36 yielded standardized planar GDS; these represented 31 physical-geometry equivalence classes and exactly 31 v4 layout vectors. The two semantic/context-only classes remain in the audit manifest. A separate supplied-layout sample contained 144 distinct vectors among 144 layouts; that full-vector collision audit is reported in the text rather than inferred from this two-dimensional atlas.

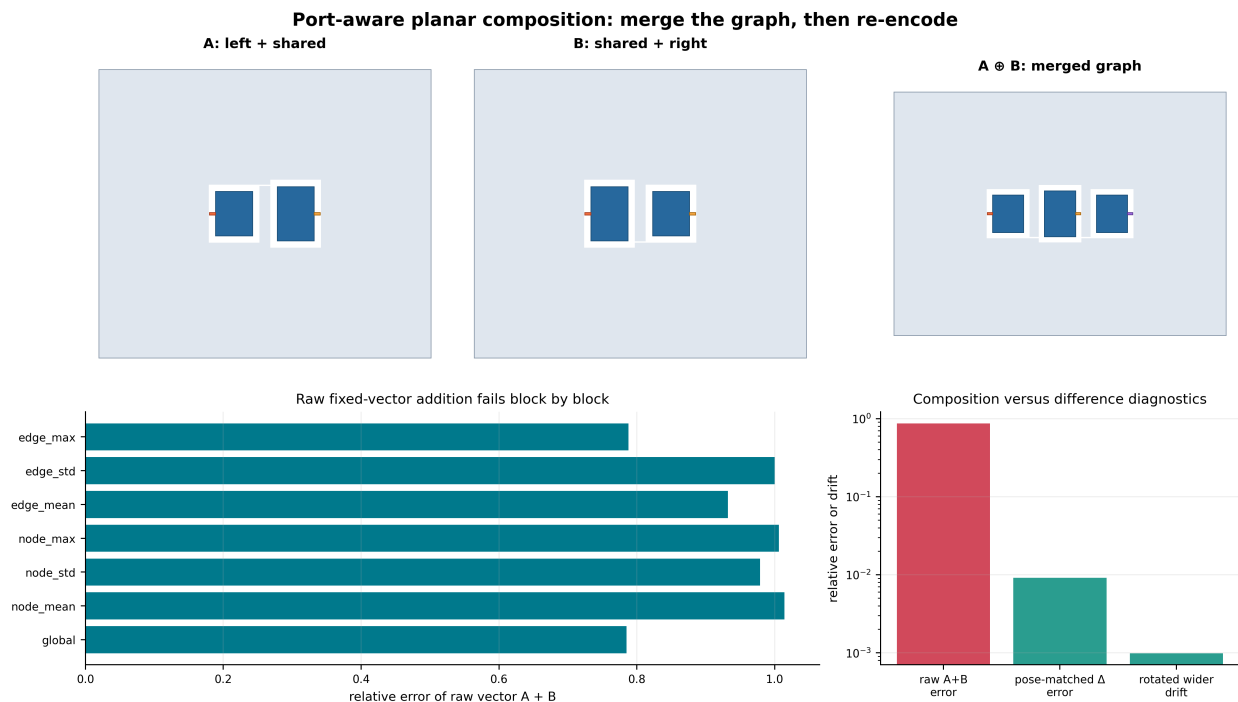


Figure 7: Controlled representation differences and port-aware planar composition. The required comparison is between a composed representation and the independently parsed joined GDS, including conductor-graph and matrix-shape agreement. Raw vector addition fails with 0.869 relative error, whereas graph composition returns three conductors plus ground and pose-matched difference directions agree to cosine 0.99997.

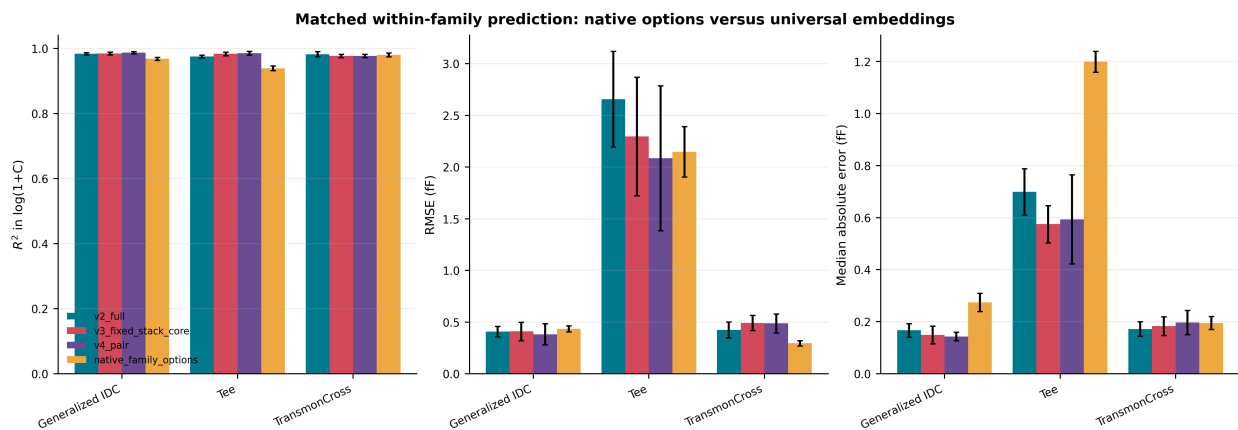
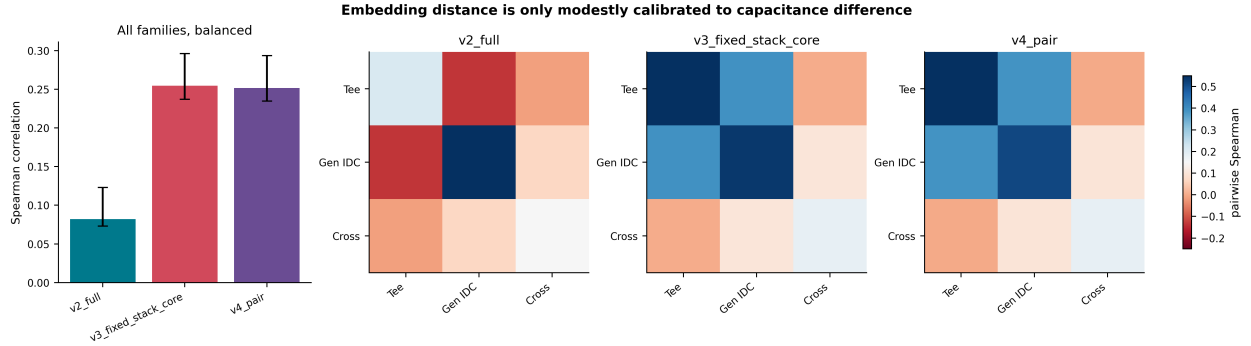


Figure 8: Matched within-family native-parameter and embedding prediction. Panels show repeated-split R^2 , RMSE, and median absolute error under one shared model protocol. v4 obtained $R^2 = 0.987$, 0.986, and 0.978; native options obtained 0.969, 0.940, and 0.981. The separate leave-one-family-out audit is reported in the text and does not establish v4 dominance on the transverse-cross target.



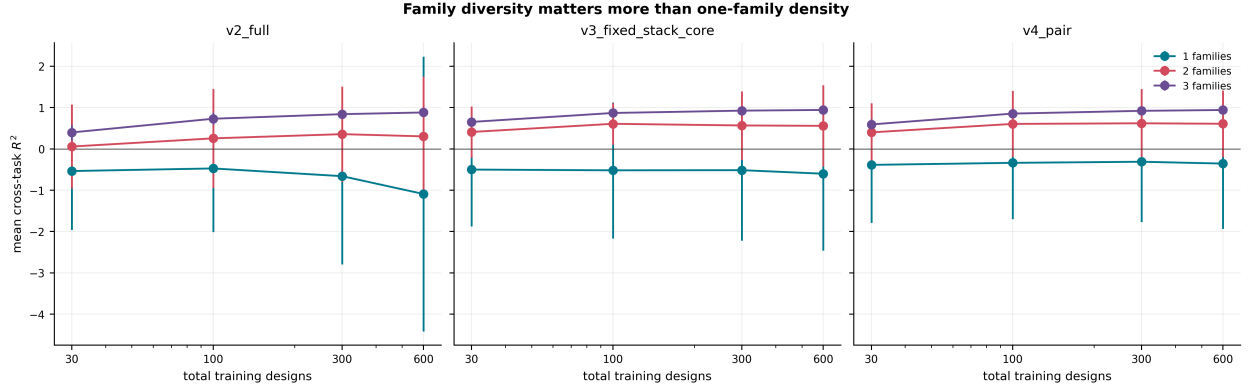


Figure 11: Controlled growth of a planar generalist model. Sample-count-matched stages separate within-family density from nearby-family, distant-family, and mixed-family diversity. At 600 designs, three-family v3/v4 training reaches mean cross-task $R^2 \approx 0.94$, whereas one-family training remains negative; topology coverage, not row count alone, controls generalization.

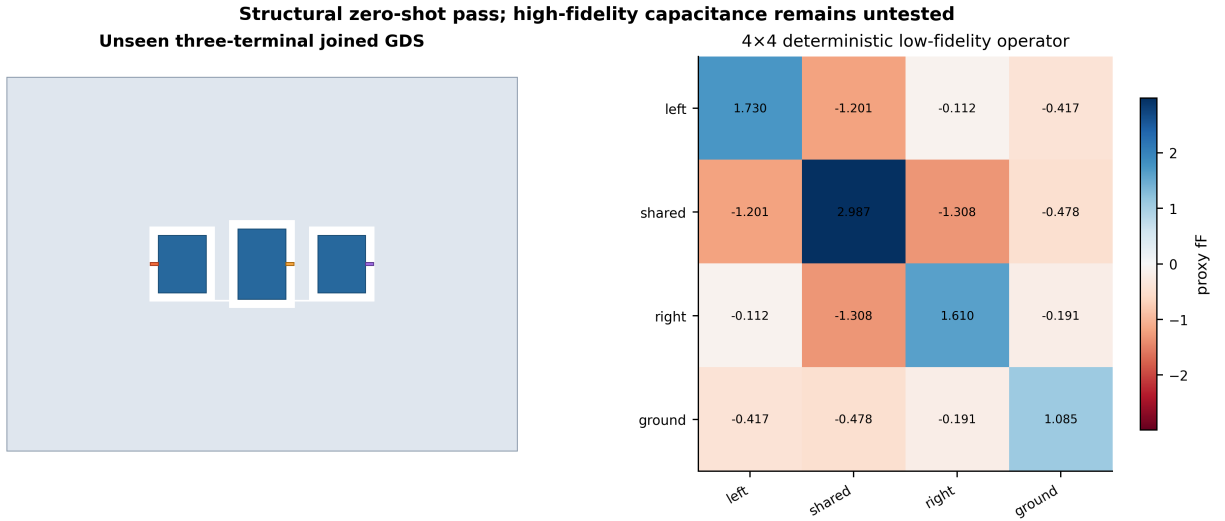


Figure 12: Unseen planar composite structural test. The standardized GDS is parsed as three signal conductors, and v4 emits the corresponding 4×4 low-fidelity conductor-plus-ground operator. This is not a high-fidelity zero-shot prediction; composite solver labels and a target-only adaptation control remain identified data-acquisition requirements.